MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCCSC202 – Introduction to Database Management System

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

(Answer all questions, each question carries 2 marks.)

1. Define a Database Management System and compare DBMS with RDBMS. [K] [CO1]
2. Explain the relationship between character, field, record, and file. [K] [CO1]
3. Write the difference between single-valued and multi-valued attributes. [U] [CO2]
4. Distinguish between a candidate key and a primary key. [U] [CO2]
5. Define a table and a view in MySQL. [K] [CO3]

[5 x 2 = 10]

Part B

Short Answer Questions

*(Answer **5** out of 7 questions, each question carries **5** marks.)*

6. Discuss the ACID properties of a DBMS. [U] [CO1]
7. Explain the role of a Database Administrator (DBA). [K] [CO1]
8. Explain the difference between one-to-one, one-to-many, and many-to-many relationships. [U] [CO2]
9. Explain how functional dependency is used in normalization and describe First, Second, and Third Normal Forms. [U] [CO2]
10. Describe different MySQL data types with examples. [U] [CO3]
11. Write MySQL queries to demonstrate the use of aggregate functions such as COUNT, SUM, and AVG. [A] [CO3]
12. Create a table EMPLOYEE with suitable fields and demonstrate the use of INSERT, UPDATE, and DELETE commands. [A] [CO3]

[5 x 5 = 25]

Part C

Essay Questions

*(Answer **1** out of 2 questions, each question carries **15** marks.)*

13. Explain the Entity–Relationship (ER) model used in database design and illustrate it using a suitable ER diagram. [U] [CO2]
14. Describe various MySQL command types used for database definition (DDL), manipulation (DML), and control (DCL). [U] [CO3]

[1 x 15 = 15]